

Dichen Li

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Research Interests

I study **cross-embodiment learning** and **reinforcement learning for robot loco-manipulation** on humanoid, quadruped, and hexapod robots. My research aims to develop agents that can learn from collective robot trajectories, generalize across diverse physical embodiments, and scale effectively with data. This forms the foundation for enabling agents to adapt to unpredictable changes such as worn motors, customized hardware, and external disturbances, without relying on precisely calibrated parameters. I am also interested in exploring robot loco-manipulation skills as a stepping stone toward broader task generalization, where diverse interaction patterns can expose the limits of scalable cross-embodiment learning.

Education

University of California, San Diego <i>Master's Program in Electrical and Computer Engineering</i> GPA: 3.96/4.00 Research interests: cross-embodiment robot learning, embodiment adaptation	09/2024-03/2026 San Diego, CA
University of California, Berkeley <i>Exchange Program in Electrical Engineering and Computer Sciences</i> Research interests: reinforcement learning for robot loco-manipulation	09/2023-06/2024 Berkeley, CA
Xi'an Jiaotong University (C9 & 985) <i>Bachelor of Engineering in Automation, Elite Program</i> GPA: 3.92/4.30 Ranking: 19/220 <i>Graduate from the highest department Qian Xuesen Honors College</i>	09/2020-06/2024 Xi'an, China

Selected Publications

- Bo Ai*, Liu Dai*, Nico Bohlinger*, **Dichen Li*** (co-first author), Tongzhou Mu, Zhanxin Wu, K. Fay, Henrik I. Christensen, Jan Peters, Hao Su.
Towards Embodiment Scaling Laws in Robot Locomotion.
Conference on Robot Learning (CoRL), 2025.
- Tomson Qu, **Dichen Li** (second author), Avidesh Zakhori, Wenhao Yu, Tingnan Zhang.
Versatile Locomotion Skills for Hexapod Robots.
International Conference on Intelligent Robots and Systems (IROS), 2024.

Awards

- Honored Graduate Students, Xi'an Jiaotong University **2024**
- Scholarship of Elite Program (Yuejie program), Xi'an Jiaotong University **2023**
- First Prize of China Undergraduate Mathematical Contest in Modeling, Shaanxi Province **2022**

Research Experiences

Embodiment Adaptation in Robot Locomotion Project <i>Graduate Research Assistant advised by Prof. Hao Su</i> Trained a locomotion policy for robots with continuously varying embodiment parameters and designed an adaptation module to infer unknown parameters and enhance robustness.	04/2025-Present SU Lab, UC San Diego
Large-scale Embodiment Scaling Law Project <i>Graduate Research Assistant advised by Prof. Hao Su</i> Developed a framework to distill locomotion policies from 1,000 diverse robot embodiments into a single generalizable policy, evaluated in Isaac Lab and deployed in real-world on Go2/H1. Discovered that training on accumulated embodiment improves performance, and that embodiment scaling outperforms data scaling.	10/2024-05/2025 SU Lab, UC San Diego
Hexapod Robot Locomotion skills with Reinforcement Learning <i>Undergraduate Research Intern advised by Prof. Avidesh Zakhori</i> Trained and distilled 3 reinforcement learning policies for hexapod loco-manipulation, and deployed them on real robots, achieving stair climbing, cave-like passages traversal (height from 22cm to 10cm), and obstacle avoidance (40cm blocks).	09/2023-05/2024 UC Berkeley
Four-Wing UAV Control and Navigation <i>Undergraduate Research Assistant</i> Developed real-time UAV flight control with multi-level PID in C/Keil; integrated indoor localization via ROS and hector_mapping; and implemented object detection in Python using YOLOv5.	04/2022-08/2023 Xi'an Jiaotong University

Skills

- Coding & Tools: Python, PyTorch, C, MATLAB, Java, Linux, Bash, LaTeX, Git, Docker, Huggingface
- Robotics & robots: ROS, ROS2, Isaac Lab, Mujoco, SLAM, Unitree GO2, Unitree H1, Raspberry Pi
- Language: English (fluent), Chinese (native)